



Recruiter Brain

AI Candidate Ranking That Thinks Like a Great Recruiter

Redrob Hackathon 2026 | TheOne

Why keyword matching fails

- Keyword filters miss Tier-5 candidates who built RAG before it had a name
- A perfect skill list means nothing if the candidate hasn't logged in for 6 months
- HR Managers with 9 AI keywords rank above ML Engineers who shipped real systems

The sample submission proves this — rank 1 is an HR Manager

What great recruiters actually do

✗ Keyword Matching

- Matches literal words, ignoring synonyms and context
- Treats all skills equally regardless of depth
- Ignores reachability and availability signals
- Easily gamed and fooled by keyword stuffers

✓ Recruiter Brain

- Reads career trajectory and progression patterns
- Weighs production experience vs pure research
- Checks availability, response rate, and notice days
- Detects and filters out impossible profiles and honeypots

5-layer pipeline

1

Data Ingestion

100K candidates + JD loaded into RAM

2

Feature Extraction

40+ signals per candidate calculated in parallel on all 100K

3

Pre-filter

Hard rules eliminate 97% of non-fits in 28 seconds

4

Deep Scoring

Semantic embeddings + 5-axis scoring computed on top 3000 survivors

5

Output

Top 100 with unique, rank-consistent natural language reasoning

5-axis scoring — not just keywords

Axis	Weight	What it measures
Semantic fit	35%	sentence-transformers cosine similarity between JD & resume summary
Hard requirements	30%	embeddings, vector DB, ranking eval, production ML, and python depth
Career trajectory	15%	product company experience, recent role relevance weights (Last role = 3x)
Availability	15%	active last 90d, open to work status, response rate, notice period
Location fit	5%	Pune/Noida exact match vs India tier-1 vs India other vs outside

We catch what others miss

5 detection rules for impossible profiles:

- Expert skill explosion: 8+ expert skills with <5 years experience
- Title-skill mismatch: Marketing Manager with 5+ AI expert skills
- Date inconsistency: role end_date before start_date
- Career duration overflow: simple duration sum > years_experience × 12 + 24
- Ghost profile: 95%+ complete, zero activity, no GitHub, 0 applications

Hard honeypots → score multiplied by 0.05 (effectively eliminated)

PERFORMANCE

Runs in 2.3 minutes on CPU — no GPU, no API calls

28s

Stage 1: pre-filter
100K → 3000

105s

Stage 2: embed +
score top 3000

0s

Stage 3: reasoning +
CSV output

Well within 5-minute constraint. 16GB RAM. Zero external API calls.

What the shortlist looks like

Rank	Score	Why
1	0.94	"7yr ML Engineer at product startup; shipped embedding-based search to 2M users; open to work, 30d notice, Pune-based"
2	0.91	"6yr applied ML; vector DB production experience at Series B; strong GitHub activity; willing to relocate to Noida"
3	0.88	"5yr NLP + retrieval background; evaluation framework experience (NDCG, A/B); concern: 90-day notice period"

Reasoning is specific, JD-connected, and honest about concerns

WHY THIS WINS

What separates us from the field

- ✓ **No API calls** — Runs completely local. Reproducible in sandboxed Docker environment, passes Stage 3 offline checks.
- ✓ **Honeypot detection** — Timeline consistency checks and expert skill explosion filters eliminate risk of disqualification by impossible profiles.
- ✓ **Specific reasoning** — Natural language output references actual candidate facts and concerns, passing Stage 4 manual review.
- ✓ **2.3 min runtime** — Consumes only 46% of the 5-minute hackathon execution budget, leaving massive headroom for future features.

The right engineer is in the dataset.

We built the system that finds them.

GitHub: <https://github.com/redrob-hackathon/recruiter-brain>

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